

InsightOS — Intelligence Report

Dataset: good_sales_dataset_400.xlsx

Data Health Score

Overall Score: **99.9/100** — Grade: **A** (Excellent)

Key Findings

- Sales and Profit are strongly correlated ($r=0.93$), indicating they may measure a similar underlying phenomenon.

Recommendations

- Explore correlation analysis between Age, Quantity, UnitPrice for deeper insights.
- Perform group-by analysis using `Region` to discover segment-level patterns.
- Conduct time series analysis on `Date` to identify temporal trends and seasonality.
- Use cluster analysis to segment records into natural groups based on numeric features.
- Review the correlation matrix to identify redundant features before reporting.

Investigations

#1: Sales and Profit are very strongly correlated (Confidence: 92.7%)

Sales and Profit have a positive correlation of 0.93. As one increases, the other reliably increases.

#2: Quantity and Sales are strongly correlated (Confidence: 65.4%)

Quantity and Sales have a positive correlation of 0.65. As one increases, the other reliably increases.

#3: UnitPrice and Sales are strongly correlated (Confidence: 64.6%)

UnitPrice and Sales have a positive correlation of 0.65. As one increases, the other reliably increases.

#4: Rare values detected in column 'OrderID' (Confidence: 90%)

Column 'OrderID' has 400 rarely occurring values (<1% frequency).

#5: Rare values detected in column 'Date' (Confidence: 90%)

Column 'Date' has 228 rarely occurring values (<1% frequency).